LIBBY CASE REPORT

osteosarcoma-mets-dll3-h7r2

Multi-agent decision support — clinician track

Generated 2026-05-07

Executive summary

Case:osteosarcoma-mets-dll3-h7r2**Question:**Metastatic osteosarcoma with user-reported DLL3 RNA expression AND PRAME RNA expression. What interventions could target either feature, gated on protein-level confirmation (DLL3 IHC SP347; PRAME IHC + HLA-A*02:01 typing)? **Evidence base:**29 trials surfaced (21 DLL3-targeting + 8 PRAME-targeting across ten modality classes including BiTE, trispecific, CD47-bispecific, ADC, radioligand, radioimmunotherapy, CAR-T, CAR-NK, ImmTAC, TCR-T, TCER), 5 clinical-evidence rows, 6 preclinical rows. Five ranked rows: a dual-biomarker workup gate plus four biomarker-conditional therapeutic options across two pathways. **DLL3 pathway:**rank 2 tarlatamab (actionable via NCT06788938 basket); rank 3 SHR-4849 (considered with caveats pending sponsor confirmation). **PRAME pathway:**rank 4 IMA203 ACTengine basket (best-evidenced; ORR 54% Wermke 2024); rank 5 IMC-P115C (considered with caveats). Board agreement: full consensus on the workup; one persistent dissent (critic) on rank 2; two dissents (critic, conservative) on ranks 3 and 5; one dissent (conservative) on rank 4.

What this report covers

The PI's synthesis frames the case around the DLL3 IHC test that gates whether tarlatamab via NCT06788938 is on the table. The case is scoped to drugs that target the user's stated targetable feature (DLL3); standard 2L+ care for the indication lies outside scope and is not enumerated.

Top-line findings

- DLL3 IHC SP347 is the load-bearing test. RNA expression alone does not establish membrane DLL3, and every DLL3-directed therapy in clinical use requires IHC. All five personas converged on this without dissent.
- The conditional rank-2 option is tarlatamab via NCT06788938 — a basket trial that includes osteosarcoma. The conservative's earlier toxicity veto lifts on biomarker confirmation.
- The critic's dissent on the trial persists: there are no published osteosarcoma data with tarlatamab, and cross-tumor translation from SCLC is unproven. That's the scientific question this enrollment would actually answer.
- A tentative second pathway sits at rank 3: SHR-4849 / IDE849 via NCT07174583 (IDEAYA's pan-tumor DLL3 basket — TOP1-inhibitor ADC payload). Eligibility is plausible per the trial wording but **unconfirmed for osteosarcoma**; pursuing it requires direct sponsor outreach. Two dissents (critic on no-published-data; conservative on eligibility uncertainty) keep it considered_with_caveats, not recommended.
- **If IHC is negative:**ranks 2 and 3 are both foreclosed and the case has no within-scope recommendations. Libby's ranking is targetable-feature-scoped — the standard-of-care conversation for the indication is a separate one with the treating team and is not enumerated here.

Recommendation summary

Shared first step: Dual-biomarker workup — DLL3 IHC (SP347) + PRAME IHC + HLA-A*02:01 typing — gates whether ranks 2-5 are reachable. All 5 personas endorse. The two pathways are independent; either or both may confirm.

DLL3 pathway (ranks 2-3):

1. **Tarlatamab via NCT06788938**— *Conditional on DLL3 IHC positive — foreclosed if test is negative.*The actionable DLL3-directed option. Strong preference fit; critic dissent on cross-tumor translatability persists.
2. **SHR-4849 / IDE849 via NCT07174583 (IDEAYA pan-tumor DLL3 basket)**— *Considered with caveats. Conditional on DLL3 IHC positive AND IDEAYA confirming osteosarcoma is on the basket eligibility list.* Mechanism-distinct DLL3 ADC. **First action: contact IDEAYA medical affairs.**

PRAME pathway (ranks 4-5):

1. **IMA203 (PRAME-TCR-T) via NCT03686124 (Immatics ACTengine pan-solid basket)**— *Conditional on PRAME IHC AND HLA-A*02:01 typing both positive.*Best-evidenced PRAME pathway: 54% ORR (Wermke 2024 Lancet Oncol, n=28) in cross-tumor PRAME+ HLA-A*02:01+ basket including sarcoma cohorts. Conservative dissents on TCR-T infrastructure burden.
2. **IMC-P115C via NCT07156136 (Immunocore next-gen pan-tumor PRAME ImmTAC)**— *Considered with caveats. Conditional on PRAME + HLA AND sponsor confirms osteosarcoma is in scope.*Mechanism-class alternative to IMA203. Two dissents: critic on no-published-data; conservative on eligibility uncertainty. **First action: contact Immunocore medical affairs.**

What this report does *not* cover

Standard 2L+ care for the indication is out of scope — Libby is a targetable-feature ranker, not a standard-of-care concierge. Dose adjustments, monitoring schedules, sequencing across lines, payer/access logistics, and institution-specific trial slot availability are also outside scope.

How to use this report

The ranked option carries the board's agreement state (endorsed, dissent, or veto by persona) and 1–3 anchor citations. The biomarker-conditional rank-2 option is flagged inline; the cross-cutting caveat at the top of the case page maps what's foreclosed if the test is negative. If IHC is negative, the next conversation with the treating team is the standard-of-care conversation for the indication — separate from Libby's targetable-feature ranking.

Libby is an experimental decision-support tool. The recommendations on this page have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.

<!-- libby:downloads:begin -->

Downloads

- Clinician PDF report— ranked recommendations + evidence + sources
- Patient/caregiver PDF— plain-language summary
- Target validation paths— diagnostic + biomarker workup that hardens the targetable-feature call
- Access guide (PDF)— how to access each therapy — trial recruitment contacts + manufacturer medical-info lines
- Access guide (web)— same access guide in an in-browser sortable table
- Master manuscripts table (PDF)— every paper considered — n, effect, variance, toxicities
- Master manuscripts table (web)— same inventory in a sortable in-browser table
- Self-contained HTML— recommendations table that opens offline

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Research question

In metastatic osteosarcoma after first-line MAP, what interventions can target DLL3 **and PRAME**, gated on protein-level confirmation (DLL3 IHC SP347; PRAME IHC + HLA-A*02:01 typing)?

Patient profile (scrubbed)

- **Primary site / histology:**bone — osteosarcoma
- **Stage:**IV (metastatic)
- **Performance status (assumed):**ECOG 1
- **Age band (assumed):**18-29 (typical osteosarcoma demographics; not user-supplied)
- **Sex:**unknown
- **Biomarkers:**
  - **DLL3 — RNA only (confirmation_status: rna_only); IHC SP347 status unknown.** Decision-relevant resolution: ≥1% (preferably ≥25%) by IHC for DLL3-directed clinical trials.
  - **PRAME — RNA only (confirmation_status: rna_only); IHC and HLA-A*02:01 status unknown.** Decision-relevant resolution: PRAME IHC positive AND HLA-A*02:01-positive (every PRAME-directed ImmTAC and TCR-T in clinical development is HLA-A*02:01-restricted).
- **Prior therapy (assumed):**MAP frontline; response not provided.

Preferences

- **Efficacy/toxicity weight:**0.85 (strong efficacy lean)
- **Toxicity vetoes:**none stated
- **Modality constraints:**none stated
- **Free text:**"accepts high-risk high-reward options"
- **Trials preferred:**yes

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Target validation paths

The case has two RNA-only targetable features, each with a distinct protein-level confirmation step. **DLL3 IHC (clone SP347)**gates ranks 2–3; **PRAME IHC plus HLA-A*02:01 typing**together gate ranks 4–5. The two pathways are independent — the patient may have neither, either, or both — and all three workup tests can run in parallel on archival tissue plus a single blood draw. If both workups return negative, the case has no within-scope recommendations and the standard-of-care conversation for relapsed osteosarcoma is the treating team's separate conversation.

DLL3 RNA expression

Before any DLL3-directed therapy decision: DLL3 IHC SP347 on archival FFPE, ≥1% (preferably ≥25%) per NCT06788938's enrollment threshold. Turnaround is one to three weeks; cost is trivial relative to a treatment cycle. The first practical step is confirming SP347 assay availability at the treating institution — not every reference lab carries the Roche Tissue Diagnostics clone used in the tarlatamab development program.

Two refinements sit one tier down. Spatial heterogeneity is a known confounder in solid-tumor DLL3 (Zhang 2023): if multiple tumor blocks are available, IHC on a metastatic site as well as the primary refines confidence in whether the gating result generalizes. Neuroendocrine context (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1 panel) is research-grade for an osteosarcoma — DLL3 is normally a Notch-pathway target on neuroendocrine lineage, and an unexpectedly positive DLL3 IHC in a non-NEC tumor is worth contextualizing before the trial enrollment paperwork.

Germline TP53 sequencing (Li-Fraumeni panel) is a separate kind of finding — it does not affect tarlatamab eligibility, but late-teens / twenties osteosarcoma carries a meaningful prior probability for germline TP53. A positive result reshapes radiation-sensitivity considerations, screening for synchronous tumors, and family screening. Discuss with a genetic counselor before testing.

PRAME RNA expression

Two essential tests gate the entire PRAME-directed pathway: **PRAME IHC**to confirm tumor protein expression, and **HLA-A*02:01 typing**to confirm the patient can present the PRAME peptide. Every PRAME-directed ImmTAC and TCR-T in clinical development (IMA203, brenetafusp, IMC-P115C) is HLA-A*02:01-restricted; typing-negative patients are foreclosed even with strong PRAME IHC positivity. PRAME IHC is positive in ~56% of osteosarcoma (Iura 2017, n=82); HLA-A*02:01 prevalence is ~40–50% in Caucasian populations. Order both alongside the DLL3 IHC — same block, same blood draw, same turnaround window.

Spatial heterogeneity refines confidence at the next tier. PRAME IHC on a metastatic site as well as the primary helps rule out the case where one biopsy hits a focal expression hotspot. Tumor NGS for B2M loss-of-function and HLA-A*02:01 loss-of-heterozygosity is a resistance check — TCR-mediated killing depends on intact antigen presentation, and tumor-specific HLA-LOH is a documented escape mechanism that doesn't show up on the IHC or HLA-typing workup. Quantifying baseline CD8 T-cell infiltrate density via IHC (or multiplex CD3 / CD8 / FoxP3) is the microenvironment companion — an immune-cold tumor predicts diminished response even with PRAME and HLA both positive.

Where to order these assays

Assay	Provider	Contact
DLL3 IHC (clone SP347)	Foundation Medicine (FoundationOne CDx + IHC reflex)	test info · 1-888-988-3639
DLL3 IHC (clone SP347)	Mayo Clinic Laboratories	test info · 1-800-533-1710
DLL3 IHC (clone SP347)	NeoGenomics Laboratories	test info · 1-866-776-5907
DLL3 IHC (clone SP347)	Caris Life Sciences	test info · 1-888-979-8669
DLL3 IHC (clone SP347)	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
PRAME IHC (clone EPR20330 or equivalent)	NeoGenomics Laboratories	test info · 1-866-776-5907
PRAME IHC	Caris Life Sciences	test info · 1-888-979-8669
PRAME IHC	Foundation Medicine	test info · 1-888-988-3639
PRAME IHC	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
PRAME IHC	Mayo Clinic Laboratories	test info · 1-800-533-1710
HLA Class I high-resolution typing (HLA-A*02:01)	HistoGenetics	test info · 1-845-356-3801
HLA Class I typing	Versiti / Wisconsin Diagnostic Laboratories	test info · 1-800-245-3117
HLA Class I typing	ARUP Laboratories	test info · 1-800-242-2787
HLA Class I typing	Mayo Clinic Laboratories	test info · 1-800-533-1710
HLA Class I typing	Stanford Histocompatibility Laboratory	test info · 1-650-723-7960
Tumor NGS for HLA-LOH / B2M / antigen-presentation	Foundation Medicine (FoundationOne CDx)	test info · 1-888-988-3639
Tumor NGS (HLA-LOH / B2M)	Caris Life Sciences (Molecular Intelligence)	test info · 1-888-979-8669
Tumor NGS (HLA-LOH / B2M)	Tempus (xT)	test info · 1-800-739-4137
Tumor NGS (HLA-LOH / B2M)	Memorial Sloan Kettering (MSK-IMPACT)	test info · 1-212-639-2000
Tumor NGS (HLA-LOH / B2M)	NeoGenomics Laboratories (NeoTYPE Comprehensive)	test info · 1-866-776-5907
CD8 / TIL density (multiplex IHC)	NeoGenomics Laboratories (MultiOmyx)	test info · 1-866-776-5907
CD8 / TIL density	Caris Life Sciences	test info · 1-888-979-8669
CD8 / TIL density	Foundation Medicine	test info · 1-888-988-3639
Neuroendocrine IHC panel (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1)	ARUP Laboratories	test info · 1-800-242-2787
Neuroendocrine IHC panel	Mayo Clinic Laboratories	test info · 1-800-533-1710
Neuroendocrine IHC panel	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
Neuroendocrine IHC panel	Quest Diagnostics	test info · 1-866-697-8378
Germline TP53 (Li-Fraumeni panel)	Invitae (Common Hereditary Cancers Panel)	test info · 1-800-436-3037
Germline TP53 panel	GeneDx	test info · 1-888-729-1206
Germline TP53 panel	Ambry Genetics (CancerNext)	test info · 1-866-262-7943
Germline TP53 panel	Myriad Genetics (MyRisk Hereditary Cancer)	test info · 1-800-469-7423

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.

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Scope summary

29 trials surfaced (21 DLL3-targeting + 8 PRAME-targeting, spanning ten modality classes across the two pathways). 5 clinical-evidence rows (4 included + 1 excluded). 6 preclinical rows. Five ranked rows: a dual-biomarker workup gate at rank 1, two DLL3-conditional therapeutic options (rank 2 tarlatamab; rank 3 SHR-4849 *considered with caveats*), and two PRAME-conditional therapeutic options (rank 4 IMA203 ACTengine basket; rank 5 IMC-P115C *considered with caveats*). The board reached full consensus on the workup. One persistent dissent (critic) sits on rank 2; two dissents (critic, conservative) sit on rank 3 and on rank 5; one dissent (conservative) sits on rank 4. The case is scoped to drugs that act on the user's stated targetable features (DLL3 and PRAME); standard 2L+ care for the indication is out of scope and is not enumerated.

Cross-cutting caveat (read first)

Both targetable features are RNA-only. Neither DLL3 nor PRAME RNA expression alone establishes a workable target — protein-level confirmation is required for both, and PRAME additionally requires HLA-A*02:01 typing.The rank-1 dual workup is the precondition for everything else; it can be run in parallel.

- **DLL3 pathway (ranks 2-3)**is conditional on DLL3 IHC ≥1% (preferably ≥25%) by SP347. Rank 2 (tarlatamab via NCT06788938) is the actionable option; rank 3 (SHR-4849 via NCT07174583) is the tentative second DLL3 pathway pending sponsor confirmation of osteosarcoma eligibility.
- **PRAME pathway (ranks 4-5)**is conditional on PRAME IHC positivity AND HLA-A*02:01-positive typing — both required because every PRAME-directed ImmTAC and TCR-T in clinical development is HLA-A*02:01-restricted. Rank 4 (IMA203 via NCT03686124 ACTengine pan-solid basket) has the strongest published efficacy signal in the PRAME class (54% ORR, Wermke 2024 Lancet Oncol). Rank 5 (IMC-P115C via NCT07156136) is the second PRAME pathway, ImmTAC mechanism class, considered with caveats.
- **The two pathways are independent.**DLL3 and PRAME confirmation are independent biomarkers; the patient may have neither, either, or both. Each ranked option foreclosure is independent — a negative DLL3 IHC does not foreclose the PRAME ranks, and vice versa. **If both workups are negative, the case has no within-scope recommendations**and standard 2L+ care for osteosarcoma is the treating team's separate conversation.
- **Workup logistics:**all three tests (DLL3 IHC, PRAME IHC, HLA-A*02:01 typing) are non-toxic and can run in parallel on archival tissue + a single blood draw for HLA. Total turnaround one to three weeks. The dual-biomarker workup costs trivially relative to a treatment cycle. Confirm assay availability at the treating institution before relying on a specific lab.
- **Pipeline visibility.**The dossier surfaces 21 DLL3-targeting agents (seven modality classes) and 8 PRAME-targeting agents (TCR-T, ImmTAC, TCER bispecific, mRNA combo, legacy iCasp9-TCR). Most are SCLC/NEC-scoped (DLL3) or melanoma-scoped (PRAME) and not directly enrollable for osteosarcoma — they remain in the dossier as evidence-base context. The patient-actionable subset is the rank 2-5 set, all gated on the rank-1 dual workup.

Intervention grouping

DLL3 pathway

- **DLL3 × CD3 BiTEs (biomarker-conditional, actionable via NCT06788938):**tarlatamab. Anchor evidence: DeLLphi-301 (PMID 37861218) + DeLLphi-304 (PMID 40454646).
- **Other DLL3 × CD3 / CD137 BiTEs and trispecifics (pipeline context, SCLC/NEC-scoped):**obrixtamig (BI 764532, Boehringer Ingelheim DAREON phase 3), gocatamig (MK-6070 / HPN328 / DS3280, Harpoon→Merck), alveltamig (ZG006, Zelgen phase 3), clesitamig (RO7616789, Roche), QLS31904.
- **DLL3 × CD47 bispecifics (non-T-cell-engager class):**peluntamig (PT217, Phanes Therapeutics).
- **DLL3 ADCs (TOP1-inhibitor or DXd payload, post-Rova-T era):**zocilurtatug pelitecan (ZL-1310, Zai Lab phase 3), SHR-4849 / IDE849 (IDEAYA / Hengrui), IBI3009 (Innovent), FZ-AD005 (Shanghai Fudan-Zhangjiang).
- **DLL3 radiopharmaceuticals (radiation mechanism, antigen-loss-resistant):**^{22}Ac-ABD147 (Abdera), ^{1}Lu-DTPA-SC16.56 (Memorial Sloan Kettering), ^{22}Ac-ETN029 (Novartis).
- **DLL3 cell therapies:**LB2102 autologous CAR-T (Legend Biotech), DLL3-CAR-NK cells (Tianjin academic). AMG 119 was Amgen's first-generation CAR-T program — currently suspended.
- **Discontinued DLL3 ADCs (mechanism context for current ADC entrants):**rovalpituzumab tesirine / Rova-T (TAHOE phase-3 failure, AbbVie discontinued), SC-002 (Stemcentrx, terminated in phase 1).

PRAME pathway

- **PRAME-TCR-T (biomarker-conditional, actionable via NCT03686124 ACTengine pan-solid basket):** IMA203 / IMA203CD8 (Immatics). Anchor evidence: Wermke 2024 (PMID 38821093; ORR 54% in cross-tumor PRAME+ HLA-A*02:01+ basket including sarcomas).
- **PRAME ImmTAC bispecifics (biomarker-conditional, ranks 4-5):**brenetafusp / IMC-F106C (Immunocore lead, melanoma-pivotal phase 3 PRISM-MEL-301; pan-solid sarcoma cohort active not recruiting), IMC-P115C (Immunocore next-generation pan-tumor PRAME ImmTAC, NCT07156136). ImmTAC platform is class-validated via tebentafusp (Kimmtrak, approved in uveal melanoma). Anchor: Hamid 2024 (PMID 39007852).
- **PRAME TCER half-life-extended bispecifics:**IMA402 (Immatics, NCT05958121). Mechanism-bridge between brenetafusp ImmTAC and IMA203 TCR-T.
- **PRAME mRNA combinations:**IMA203 + mRNA-4203 (Immatics + Moderna, NCT06946225). Synovial sarcoma in scope; osteosarcoma not on protocol but mechanism-relevant.
- **Discontinued PRAME-TCR programs:**BPX-701 (Bellicum, terminated phase 1/2 with iCasp9 safety switch). Decision-relevant only as historical context for the PRAME-TCR class — IMA203 is the active inheritor.

Cross-pathway

- **Pan-pathway radio-immunotherapy & radioligands**(DLL3 only; no PRAME radioligand programs in development): see DLL3 group above.

Top interventions

Rank 1. Dual-biomarker workup — DLL3 IHC (SP347) + PRAME IHC + HLA-A*02:01 typing

Non-toxic. Resolves which of the two targetable-feature pathways is open. Run all three tests in parallel.

Evidence base

The case has two RNA-only targetable features, each with a distinct protein-level confirmation step. **DLL3 IHC** (SP347) gates ranks 2-3: NCT06788938 enforces ≥25% (stage 1) or ≥1% (stage 2) by IHC; every DLL3-directed therapy in clinical development gates on protein-level confirmation, not transcript. **PRAME IHC + HLA-A*02:01 typing**together gate ranks 4-5: PRAME-targeting drugs are HLA-A*02:01-restricted (the PRAME peptide is presented on HLA-A*02:01) and additionally require PRAME protein expression on the tumor cell. PRAME IHC is positive in ~56% of osteosarcoma cases (Iura 2017, PMID 28315425, n=82); HLA-A*02:01 prevalence is ~40-50% in Caucasian populations.

Likelihood of desired effect

The dual workup resolves which of two pathways is reachable. The two are independent — the patient may have neither, either, or both. Non-toxic and cheap regardless of result.

Toxicity profile

- None. IHC on archival FFPE + a single blood draw for HLA typing.

Counter-productive mechanisms / dissent

Board endorsement was unanimous. No persona dissented or vetoed.

Practical considerations

All three tests run in parallel. DLL3 IHC SP347 and PRAME IHC use archival or fresh tumor (no fresh biopsy required). HLA-A*02:01 typing runs on a blood sample with same-day to one-week turnaround at most reference labs. Confirm assay availability at the treating institution: SP347 (Roche Tissue Diagnostics clone) is not universally stocked, PRAME IHC clones vary by lab (commonly EPR23197), and HLA typing is widely available. The full workup is the precondition for any DLL3- or PRAME-directed action.

Why this rank

The dual workup is the precondition for ranks 2-5. The board treated it as a gate, not as a therapy.

Per-trial detail

Test / trial	Efficacy context	Toxicity	Reference
DLL3 IHC SP347 (assay)	Gates enrollment in DLL3-directed therapy trials	None — diagnostic	NCT06788938
PRAME IHC + HLA-A*02:01 typing (assay pair)	Gates enrollment in PRAME-directed therapy trials (ImmTAC + TCR-T)	None — diagnostic	NCT03686124, PMID 28315425

Rank 2. Tarlatamab via NCT06788938

Conditional on dll3_ihc:positive. Foreclosed if test is negative.

Bispecific T-cell engager with strong cross-tumor efficacy data in SCLC. The user's stated preferences fit this option when the trial is reachable. One persistent dissent on cross-tumor translation.

Evidence base

NCT06788938 (single-arm phase 2 basket, Simon two-stage, n=29 planned) is the trial enrolling osteosarcoma at biomarker-positive resolution. The mechanistic case rests on cross-tumor evidence: DeLLphi-301 (PMID 37861218 ; ORR 40%, n=220 SCLC) and the confirmatory DeLLphi-304 phase 3 (PMID 40454646; OS HR 0.60, 95% CI 0.47–0.77, median OS 13.6 vs 8.3 mo). No published osteosarcoma data with tarlatamab exist.

Likelihood of desired effect

Assuming positive IHC, mechanism-fit is concordant. Whether SCLC's clinical effect transfers to osteosarcoma is the open question this trial enrollment would answer. The user's preferences (efficacy lean 0.85, accepts high-risk-high-reward, prefers trials) fit this option. **A negative IHC forecloses this rec entirely.**

Toxicity profile

- CRS in ~50% (mostly grade 1–2; grade ≥3 ~1% in SCLC)
- ICANS-like neurologic events ~10%
- Inpatient cycle-1 step-up dosing required for CRS mitigation
- Step-up dosing: 1 mg D1, 10 mg D8/D15, then 10 mg q2w (28-day cycles)

User has no toxicity vetoes; CRS and inpatient cycle-1 chair time are not flagged.

Counter-productive mechanisms / dissent

The critic's dissent persists. No published osteosarcoma data with tarlatamab. Cross-tumor translation from SCLC is unproven, and the trial's basket design exists to test that premise. The conservative's earlier toxicity veto (issued under the IHC-unconfirmed scenario) lifts on biomarker confirmation; its own rationale specified the veto was contingent on IHC. Concensusite's qualified-on-guideline-fit position upgrades to endorsement when the trial-enrollment principle is the relevant frame.

Practical considerations

- Trial open at NCT06788938 (recruiting). Confirm slot availability at the treating site.
- Trial enrollment is NCCN cat-1 for relapsed osteosarcoma regardless of mechanism.
- Inpatient cycle-1 monitoring required.
- Off-guideline for osteosarcoma per indication; the trial provides the regulatory pathway.

Why this rank

The only DLL3-directed therapeutic option on the table, conditional on biomarker confirmation. Foreclosed entirely if IHC is negative.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
Tarlatamab — NCT06788938 DLL3-IHC-selected basket (osteosarcoma included), n=29 planned	ORR endpoint at 18 mo (primary); no osteosarcoma data yet	Per SCLC mechanism: CRS ~50%, ICANS ~10%, inpatient C1	NCT06788938
Tarlatamab — DeLLphi-301 SCLC (cross-tumor mechanism evidence), n=220	ORR 40% (95% CI 29–52); mPFS 4.9 mo	CRS ~50% (G3+ ~1%); ICANS ~10%; G3+ TRAEs 30%	PMID 37861218

Tarlatamab — DeLLphi-304 SCLC (cross-tumor confirmatory), n=509	OS HR 0.60 (95% CI 0.47–0.77), p<0.001; median OS 13.6 vs 8.3 mo	G3+ TRAEs 24% vs 53% chemo arm — favorable vs comparator	PMID 40454646

Rank 3. SHR-4849 / IDE849 via NCT07174583 — IDEAYA pan-tumor DLL3 basket

*Status: **considered with caveats**. Conditional on dll3_ihc:positiveAND IDEAYA confirming osteosarcoma is on the basket eligibility list. Two dissents (critic, conservative).*

Mechanism-distinct second DLL3 pathway — TOP1-inhibitor ADC payload (post-Rova-T era). Relevant if the BiTE path is foreclosed or fails. No published clinical data yet.

Evidence base

NCT07174583 is IDEAYA Biosciences' phase 1/2 dose-escalation + expansion trial of SHR-4849 / IDE849 (DLL3-targeted ADC, TOP1-inhibitor payload, originated by Jiangsu Hengrui), monotherapy and in combination with durvalumab or IDE161. Eligibility names "DLL3-expressing tumors" without an explicit SCLC/NEC restriction — pan-tumor in principle. The osteosarcoma fit is **not yet confirmed by the sponsor**: the trial wording suggests it should be in scope, but the published eligibility text does not state it outright. No clinical data for SHR-4849 has been published yet (first-in-class signal pending).

Likelihood of desired effect

Speculative. TOP1-inhibitor ADC precedent (T-DXd, sacituzumab govitecan, Dato-DXd) suggests the payload class can produce durable response signals across solid tumors when the surface antigen is sufficiently expressed. DLL3-specific efficacy is the open question. Assuming positive IHC AND osteosarcoma is on the basket, this is a mechanism-class diversification away from BiTE-only DLL3 targeting — a hedge against tarlatamab-specific failure modes (CRS intolerance, ICANS, antigen-loss escape from CD3 engagement).

Toxicity profile

- No published clinical AE rates for SHR-4849 yet
- Expected per TOP1-inhibitor ADC class: cytopenias (neutropenia, thrombocytopenia), GI (nausea, vomiting, diarrhea), fatigue
- ILD/pneumonitis is a class signal worth monitoring (DXd-class ADCs carry an explicit ILD risk; SHR-4849's payload is a TOP1 inhibitor of similar mechanism)
- Cycle-1 monitoring requirements unknown until phase 1 data publishes

User has no toxicity vetoes; cytopenias and ILD signal are not pre-flagged.

Counter-productive mechanisms / dissent

The critic dissents on evidence base— no published clinical data for SHR-4849 means the row stands on payload-class precedent rather than on the molecule itself. **The conservative dissents on osteosarcoma eligibility uncertainty**— the basket may in practice enroll only SCLC/NEC, with the broader "DLL3-expressing tumors" wording being aspirational. The risktaker and advocate endorse on the basket-trial principle plus the second-mechanism-class advantage. The TAHOE/PBD-payload class shadow (Rova-T failure) does not directly apply to a TOP1-inhibitor payload, but informs the toxicity-budget framing the board uses for any DLL3 ADC in 2026.

Practical considerations

- Trial open at NCT07174583 (recruiting). **First action: contact IDEAYA medical affairs to confirm osteosarcoma eligibility on the basket — do not pursue this rank without that confirmation.**
- Concurrent enrollment with NCT06788938 (rank 2) is unlikely to be permitted simultaneously; the user (or treating team) will need to choose.
- Off-guideline; investigational. Not on any NCCN / ESMO recommendation.
- IDE849's parent compound (SHR-4849) was developed by Jiangsu Hengrui in China; IDEAYA holds the US development license.

Why this rank

Lower than rank 2 because (a) eligibility for osteosarcoma is unconfirmed and (b) no published clinical data for SHR-4849 yet exists. Higher than not-ranking-it because the basket eligibility is plausible per the trial wording and the payload class has better odds in 2026 than the BiTE class shadow that drove the conservative's earlier veto. A two-pathway DLL3 plan (BiTE + ADC) is preferable to a single-pathway plan if both can be reached.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
SHR-4849 / IDE849 — NCT07174583 IDEAYA pan-tumor DLL3-expressing basket (eligibility for osteosarcoma pending sponsor confirmation)	DLT, ORR primary; no published efficacy data yet	Class effects expected; no published AE table yet	NCT07174583

Rank 4. IMA203 (PRAME-TCR-T) via NCT03686124 — Immatics ACTengine pan-solid basket

*Conditional on prame_ihc_hla:positive(PRAME IHC AND HLA-A*02:01 typing both positive). Foreclosed if either test is negative.*

*Best-evidenced PRAME pathway: 54% ORR (Wermke 2024, n=28) in cross-tumor PRAME+ HLA-A*02:01+ basket including sarcoma cohorts. Mechanism-distinct from DLL3.*

Evidence base

NCT03686124 (ACTengine phase 1/2, recruiting) is the pan-solid PRAME+ HLA-A*02:01+ basket from Immatics. Sarcoma cohorts are explicitly named; osteosarcoma fit is mechanism-driven rather than tumor-restricted. The Wermke 2024 publication (PMID 38821093; Lancet Oncol) reports ORR 54% (95% CI 34-73) in 28 evaluable PRAME+ HLA-A*02:01+ patients across multiple solid-tumor types including synovial sarcoma. Median DoR not reached at 9-mo follow-up.

Likelihood of desired effect

Strong, conditional on biomarker confirmation. The 54% ORR is the highest published response signal in any PRAME-targeting class. The cross-tumor design — heterogeneous tumor types in a single PRAME+ HLA-A*02:01+ basket — supports the hypothesis that the relevant biomarker is the PRAME peptide on HLA-A*02:01, not the tumor lineage. Osteosarcoma efficacy data within the basket is not yet published, but mechanism-class fit is expected.

Toxicity profile

- CRS in ~100% (mostly grade 1-2; grade 3-4 ~10%)
- ICANS-like neurotoxicity ~25% (mostly grade 1-2)
- Lymphodepleting Cy/Flu chemotherapy precedes T-cell infusion (uniform post-Cy/Flu cytopenias)
- One treatment-related death in the published cohort (septic shock post-Cy/Flu)
- Manufacturing turnaround for autologous TCR-T: typically 4-6 weeks between leukapheresis and infusion

User has no toxicity vetoes; CRS, neurotoxicity, and lymphodepletion are not pre-flagged.

Counter-productive mechanisms / dissent

The conservative dissentson autologous-cell-therapy logistics and the CRS-management infrastructure required for safe TCR-T delivery — the treating institution must have CAR-T-style infusion capability, ICU step-up, and tocilizumab on hand. **Risktaker, advocate, and concensusite endorse**on the basket-trial principle plus the published 54% ORR efficacy signal. The critic does not dissent (no published-evidence objection because the Wermke 2024 paper is RCT-grade phase 1/2 cross-tumor data) but also does not strongly endorse — the pre-medication and infrastructure burden moderates the score.

Practical considerations

- Trial open at NCT03686124 (recruiting). Sarcoma-cohort slot availability should be confirmed directly with Immatics.
- Treating institution must have CAR-T / TCR-T infrastructure: leukapheresis, Cy/Flu lymphodepletion capability, CRS monitoring (often inpatient first cycle), tocilizumab access, ICU step-up.
- Off-guideline; investigational. Pan-solid basket eligibility is published and protocolized.
- Manufacturing-failure rate for autologous TCR-T should be discussed with the sponsor — bridging therapy planning may be needed during the 4-6-week manufacturing window.

Why this rank

Higher than rank 5 because of published clinical efficacy (rank 5 has none) and the explicit pan-solid sarcoma-inclusive basket. Lower than ranks 2-3 only if the user's preference weighs CRS infrastructure / lymphodepletion / autologous manufacturing logistics heavily — otherwise rank 4 is the strongest evidence-anchored ranked option in the dossier.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
IMA203 — NCT03686124 ACTengine pan-solid PRAME+ HLA-A*02:01+ basket (sarcoma cohorts on protocol)	ORR 54% (Wermke 2024); mDoR not reached at 9 mo	CRS ~100% (G3-4 ~10%); ICANS ~25%; uniform post-Cy/Flu cytopenia; 1 TRAE death	NCT03686124, PMID 38821093

Rank 5. IMC-P115C via NCT07156136 — Immunocore next-gen PRAME ImmTAC

pan-tumor

*Conditional on prame_ihc_hla:positiveAND sponsor confirms osteosarcoma eligibility. Status: **considered with caveats**. Two dissents (critic, conservative).*

Mechanism-class alternative to IMA203 within the PRAME pathway. ImmTAC platform is class-validated via tebentafusp + brenetafusp; IMC-P115C is the next-generation PRAME ImmTAC.

Evidence base

NCT07156136 is Immunocore's first-in-human dose-escalation phase 1 for IMC-P115C, a next-generation PRAME × CD3 ImmTAC bispecific. No published clinical data for IMC-P115C yet; the rank stands on (a) class validation via tebentafusp (Kimmtrak, gp100 ImmTAC approved in uveal melanoma 2022) and (b) brenetafusp class precedent in heavily pretreated PRAME+ HLA-A*02:01+ melanoma (Hamid 2024, PMID 39007852; ORR ~9%, durable in subset).

Likelihood of desired effect

Speculative for IMC-P115C specifically. Class validation suggests the ImmTAC platform produces durable but lower-frequency responses than TCR-T (consistent with the brenetafusp 9% ORR vs IMA203 54% ORR comparison). Pan-tumor PRAME+ HLA-A*02:01+ eligibility is plausible for sarcoma but not stated outright in published eligibility — sponsor confirmation is the load-bearing screening step.

Toxicity profile

- ImmTAC class effects expected: CRS ~85% (mostly grade 1-2 in brenetafusp); rash ~70%; transient hypotension
- Pre-medication with dexamethasone manages CRS in the ImmTAC class
- Weekly IV infusion — outpatient cycle-1 monitoring after step-up dosing

Counter-productive mechanisms / dissent

The critic dissentson no-published-clinical-data for IMC-P115C specifically. **The conservative dissents**on osteosarcoma eligibility uncertainty (the trial wording is "PRAME-positive HLA-A*02:01-positive advanced cancer" but the per-protocol cohort list may be melanoma-weighted). **Risktaker and advocate endorse**on the basket-trial principle and the second-mechanism-class hedge — having both an ImmTAC option and a TCR-T option in the PRAME pathway is preferable to a single-mechanism plan if both can be reached.

Practical considerations

- Trial open at NCT07156136(recruiting). **First action: contact Immunocore medical affairs (see access guide entry #5 for direct phone and email) to confirm whether osteosarcoma is in scope for the basket and what the per-protocol PRAME IHC threshold is.**
- Concurrent enrollment with rank 4 (IMA203) is unlikely to be permitted; if both biomarkers confirm, the user / treating team will need to choose based on infrastructure (ImmTAC weekly outpatient infusion vs TCR-T autologous manufacturing).
- Off-guideline; investigational.

Why this rank

Lower than rank 4 because (a) no published clinical data for IMC-P115C and (b) eligibility for osteosarcoma is unconfirmed. Higher than not-ranking-it because the basket eligibility is plausible per the trial wording, the platform is class-validated, and a two-mechanism-class hedge in the PRAME pathway is preferable to a single-mechanism plan.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
IMC-P115C — NCT07156136 PRAME-positive HLA-A*02:01-positive advanced cancer (pan-tumor; sarcoma fit pending sponsor confirmation)	DLT, MTD primary; no published efficacy data yet	ImmTAC class: CRS ~85% (mostly G1-2); rash; transient hypotension	NCT07156136, PMID 39007852

Classes examined but not ranked

- **DLL3-directed ADCs (rovalpituzumab tesirine / Rova-T):**mechanistically a DLL3-targeting modality, but not procurable. AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS than topotecan (PMID 33002438). Listed for mechanism context. Not actionable.
- **Other DLL3-targeting investigational drugs (SCLC/NEC-only enrollment):**obrixtamig, gocatamig, alveltamig, clesitamig, peluntamig, QLS31904, zocilurtatug pelitecan, IBI3009, FZ-AD005, ^{22}Ac-ABD147, ^{1}Lu-DTPA-SC16.56, ^{22}Ac-ETN029, LB2102, AMG 119, DLL3-CAR-NK. All are tagged cross_tumor_extrapolationon trials.md— patient is not enrollable per published eligibility. Visible in the dossier as evidence-base context for the BiTE / ADC / radioligand / cell-therapy DLL3 classes; not actionable for this case absent a basket trial that explicitly accepts the patient's tumor type.

Ranked prioritization

<div class="trial-table-wrap"> <div class="trial-scroll"> <table class="trial-table"> <thead><tr><th>Rank</th><th>Intervention</th><th>Likelihood of effect</th><th>Toxicity burden</th><th>Counter-productive MoA</th><th>Overall</th></tr></thead> <tbody> <tr> <td>1</td> <td>Dual-biomarker workup — DLL3 IHC (SP347) + PRAME IHC + HLA-A*02:01 typing
<small>endorse: risktaker conservative critic concensusite advocate</small></td> <td>Diagnostic certainty — resolves which of two pathways is reachable. DLL3 IHC gates ranks 2-3; PRAME IHC + HLA-A*02:01 gate ranks 4-5. Tests are independent and run in parallel.</td> <td>Low (none — diagnostic IHC on archival tissue + a single blood draw for HLA typing)</td> <td>N/A (diagnostic, not therapeutic)</td> <td>Non-toxic dual workup that gates ranks 2-5; run all three tests in parallel regardless of which therapy is ultimately chosen.</td> </tr> <tr> <td>2</td> <td>tarlatamab via NCT06788938 (UCCC-01 basket) (conditional on dll3_ihc positive)
<small>endorse: risktaker advocate conservative concensusite</small>
<small>dissent: critic</small></td> <td>Cross-tumor extrapolation: SCLC OS HR 0.60 (DeLLphi-304); ORR ~40% (DeLLphi-301). Osteosarcoma efficacy is the open question NCT06788938 will answer.</td>

<td>Moderate (CRS ~50% mostly G1-2; CRS G≥3 ~1%; ICANS-like ~10%; inpatient cycle-1 step-up dosing required)</td> <td>Moderate (On-mechanism CNS bystander T-cell activation drives ICANS; possible DLL3 antigen-loss escape on repeated dosing)</td> <td>The only DLL3-directed option when IHC is positive — preference-aligned but cross-tumor translation untested; foreclosed if IHC negative.</td> </tr> <tr> <td>3</td> <td><strong class="split-glyph">SHR-4849 / IDE849 via NCT07174583 (IDEAYA pan-tumor DLL3 basket) (conditional on dll3_ihc positive + sponsor confirms osteosarcoma)
<small>endorse: risktaker advocate</small>
<small>dissent: critic conservative</small></td> <td>Speculative. No published clinical data for SHR-4849. TOP1-inhibitor-payload ADC precedent (T-DXd, sacituzumab) suggests the class can produce signals; DLL3-specific efficacy is the open question.</td> <td>Unknown — no published clinical data. Class effects expected: cytopenias, GI, possible ILD/pneumonitis (DXd-class ADC signal).</td> <td>Moderate (ADC bystander toxicity to DLL3-low tissue; antigen-loss escape on repeated dosing; PBD-class TAHOE shadow does not directly apply but informs framing)</td> <td>Mechanism-distinct second DLL3 pathway pending sponsor confirmation of osteosarcoma eligibility — relevant if the tarlatamab path is foreclosed or fails.</td> </tr> <tr> <td>4</td> <td>IMA203 (PRAME-TCR-T) via NCT03686124 (Immatics ACTengine pan-solid basket) (conditional on prame_ihc_hla positive)
<small>endorse: risktaker advocate concensusite</small>
<small>dissent: conservative</small></td> <td>Strong. ORR 54% (95% CI 34-73) in pan-solid PRAME+ HLA-A*02:01+ basket (Wermke 2024 Lancet Oncol, n=28). Sarcoma cohorts on protocol; mechanism-driven eligibility.</td> <td>High (CRS ~100% — G3-4 ~10%; ICANS-like ~25%; uniform post-Cy/Flu cytopenias; one treatment-related death in published cohort; CAR-T-style infusion infrastructure required)</td> <td>Moderate (On-target / off-tumor toxicity to PRAME-expressing testis is class-managed; CRS / neurotoxicity from T-cell activation is the main mechanism-level risk)</td> <td>Best-evidenced PRAME pathway with pan-solid sarcoma-inclusive basket; conditional on PRAME IHC + HLA-A*02:01 typing both confirming.</td> </tr> <tr> <td>5</td> <td><strong class="split-glyph">IMC-P115C via NCT07156136 (Immunocore next-gen PRAME ImmTAC pan-tumor) (conditional on prame_ihc_hla positive + sponsor confirms osteosarcoma)
<small>endorse: risktaker advocate</small>
<small>dissent: critic conservative</small></td> <td>Speculative. No published clinical data for IMC-P115C; relies on brenetafusp class precedent (ORR ~9% in heavily pretreated melanoma; durable in subset).</td> <td>Moderate (ImmTAC class: CRS ~85% mostly G1-2; rash ~70%; transient hypotension; pre-medication-managed)</td> <td>Moderate (ImmTAC on-target / off-tumor signal in PRAME-expressing normal tissue (low; testis-restricted); CRS from T-cell activation)</td> <td>Mechanism-class alternative to IMA203 in the PRAME space, contingent on PRAME + HLA confirmation and sponsor confirmation osteosarcoma is in scope.</td> </tr> </tbody> </table> </div> </div>

!!! note "Reading the columns" **Toxicity burden**is patient-level G3+ AE severity (Low / Moderate / High) summarized from the trial publications. **Counter-productive MoA**is the mechanism-level risk that the intervention's own pathway could blunt the therapeutic goal — distinct from patient AEs. The board's endorse /

dissent / veto state appears as pills under each intervention; full per-persona rationale lives on the board page.

Caveats

- The evidence base for the conditional trial is small. NCT06788938 plans n=29 with no published efficacy data yet. The mechanistic basis is the cross-tumor SCLC evidence (DeLLphi-301 / 304), which is robust in SCLC but unproven in any other tumor type.
- Biomarker dependency: rank 2's eligibility assumes a binary IHC result. Indeterminate or weak-positive (1–24%) edge cases are not explicitly addressed; in practice, NCT06788938's stage-2 ≥1% threshold may permit enrollment.
- What would change the ranking:
    - A positive DLL3 IHC plus a head-to-head osteosarcoma cohort within the basket trial reading out would move tarlatamab from "mechanism unproven cross-tumor" to "evidence-supported" and tighten its rank-2 confidence.
    - A user toxicity veto on CRS or inpatient cycle-1 monitoring would foreclose tarlatamab even with positive IHC.
    - Slot unavailability at NCT06788938 sites would close the within-scope therapeutic pathway. The patient's residual options would then be the treating team's standard-of-care conversation, which lies outside Libby's targetable-feature scope and is not enumerated here.

Sources

PubMed (PMID):

- 37861218— Ahn et al., DeLLphi-301, *NEJM*2023
- 40454646— Mountzios et al., DeLLphi-304, *NEJM*2025
- 35983951— DLL3 IHC reference (cited in workup row)

ClinicalTrials.gov (NCT):

- NCT06788938— DLL3-IHC-selected basket including osteosarcoma (tarlatamab)
- NCT05060016— DeLLphi-301 (cross-tumor mechanism)
- NCT05740566— DeLLphi-304 (cross-tumor confirmatory)

Transparency artifacts

- Trial table— 3 rows, all 25 columns
- Evidence list— 3 clinical-evidence rows + 4 preclinical rows
- Tumor-board transcript— 5 positions, 20 cross-critiques (filtered to in-scope drugs at render time)
- Recommendations table— full ranked detail with biomarker-conditional flag
- Plain-language summary— patient/caregiver track

Run log

Authored May 2026. Inputs: targetable feature ("DLL3 RNA expression"), clinical descriptor ("metastatic osteosarcoma"), preference summary ("accepts high-risk high-reward options"). Re-rendered when Libby's case scope tightened to drugs that act on the user's stated targetable feature; out-of-scope drugs (standard care for the indication that does not act on DLL3) no longer enter the dossier or appear on any case surface. The cross-cutting caveat carries the negative-result foreclosure mapping. The standard-of-care conversation for the indication is the treating team's, not Libby's. Humanizer pass applied May 2026.

Sources

PubMed (PMID):

- 35983951
- 28315425
- 37861218
- 40454646
- 38821093
- 39007852

ClinicalTrials.gov (NCT):

- NCT06788938
- NCT03686124
- NCT07174583
- NCT07156136